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DFIT Based Empirical Approach to Predict Unconventional Tight Well Productivity Index and Post Closure Formation Related Properties

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Abstract

Diagnostic fracture injection test (DFIT) is a widely used tool to derive key reservoir properties for unconventional reservoirs. This paper proposes a novel and simplified approach to estimate the reservoir transmissibility/permeability from DFIT closure data at linear flow, and without the need for radial flow. The work was implemented in multiple tight sand formation wells. Results yielded an estimate of transmissibility/permeability, and optimize DFIT post-injection monitoring time.

The proposed approach incorporates Closure Analysis and After Closure Analysis (ACA) results from previous DFIT tests to derive a correlation for estimating formation permeability. The approach begins by estimating injected fluid leak-off efficiency through the application of the G-function, followed by utilizing ACA to obtain reservoir transmissibility and pore pressure. The concept is built on fluid diffusivity mechanisms. During fracture closure, the fluid leak off mechanism into the formation is governed by the magnitude of Instantaneous Shut-in Pressure (ISIP), until closure pressure is reached. After closure, fluid diffusivity is influenced by another pressure mechanism that is the excess pressure as it diffuses into the formation and approach the reservoir pore pressure. Finally, the fluid efficiency from Closure Analysis is considered as a formation transmissibility index, which is then compiled with the actual transmissibility estimated from ACA to develop the prediction correlation.

In this study, the approach was applied in multiple wells where DFIT was concluded before reaching the radial flow. Additionally, the built correlation was used to evaluate the transmissibility in other wells where only closure analysis was available. The correlation developed between the fluid leak-off efficiency and reservoir transmissibility from ACA showed an exponential trend with high accuracy, with a Coefficient of Determination higher than 95%. The result of the correlation was linked to well performance using productivity index as a proxy, yielding accurate result, indicating the critical role formation transmissibility has on fluid deliverability. Applying the productivity index and the predicted formation transmissibility from the correlation matched with high accuracy which validates the methodology.

For extremely tight reservoirs, radial flow might be reached after several days, weeks, or months of monitoring pressure decline. The proposed approach saves significant DFIT post-injection time required to observe radial flow. Also, it can be used as validation tool for closure pressure picking where in some case the closure estimation utilizing G-function is selective.

Introduction

Diagnostic fracture injection test (DFIT) is a well testing tool commonly used in unconventional tight reservoirs to derive key fracture parameters and reservoir properties. Unconventional reservoirs are generally characterized as extremely tight reservoirs with a low to ultra-low permeability range. Due to the tightness of these reservoirs, DFIT is used as a reliable tool to predict essential reservoir parameters and replace conventional well testing methods of pressure buildup/drawdown. DFIT is an analogous tool to the pressure fall-off used in conventional reservoirs, where pressure decline is recorded soon after the conclusion of injection. The DFIT operation starts by injecting a small volume of clean fluid, such as treated water, into the formation to breakdown the rock and generate a minor hydraulic fracture. Once the fracture is generated, pumping is concluded and pressure decline is monitored. The general idea of DFIT is to interpret both fracture and reservoir parameters by evaluating the pressure decline of a small fracture as injected fluid slowly leaks into the formation until the fracture is closed.

Figure 1 captures a generic DFIT operation and interpretation, where there are 3 main phases of interest for detailed analysis. The most crucial phase takes place during DFIT pumping while generating a fracture, which reflects the most fluid leak-off efficiency. Then, the interpretations of the pressure decline are conducted based on pressure behavior observed in two major phases, pre-closure and post-closure. The pre-closure phase starts immediately after shutting in the well until pressure reaches the fracture closure pressure, and during this phase, the injected fluid diffuses into the formation. Interpretation of pressure transient during the pre-closure phase up to when closure occurred is referred to as the Closure Analysis (CA). CA can be conducted once fracture closure is confirmed, which is observed with the log-log diagnostic plot when the pressure transient is approaching the formation linear flow regime. Fracture parameters are determined, including closure time, closure pressure, net pressure, fluid loss coefficient, and fluid leak-off efficiency, with are all crucial to the design of hydraulic fracturing.

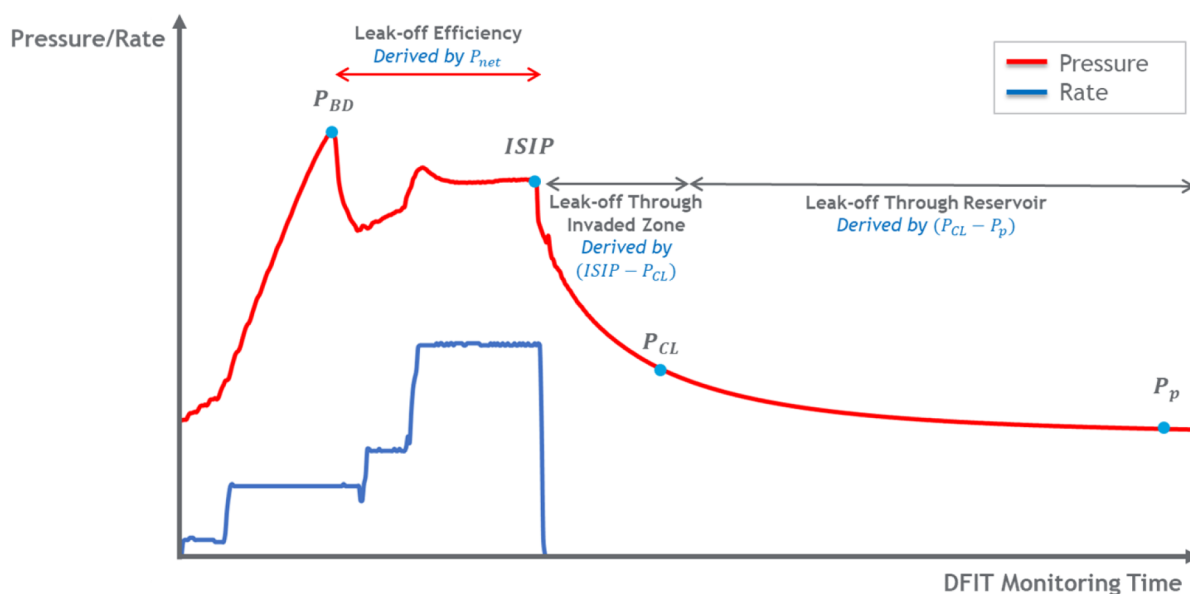


Figure 1—Example of DFIT Raw Data Showing Pressure Stages.

The post-closure phase starts soon after fracture closure pressure is reached, as pressure continues to decline with a trend that transitions from linear to radial flow, indicating true formation contribution. During this phase, the pressure charge of the surrounding region governs the leak-off until the declining pressure reaches the reservoir pore pressure. Once formation radial flow is developed, After Closure Analysis (ACA) is interpreted. The ACA defines valuable reservoir properties, including reservoir pore pressure and formation transmissibility.

Although DFIT is designed for tight reservoirs, extreme cases of low permeability and low diffusivity require a long shut-in time for radial flow to be identified, making it a bottleneck to operational efficiency. In most DFIT cases, the linear flow regime can be developed within relatively short durations, which makes CA data easily obtainable. Utilizing available CA and ACA data from previous DFITs on wells targeting the same formation to construct a tool that predicts ACA radial flow presents a valuable opportunity. This paper proposes a simplified approach to construct a tool that estimates the reservoir transmissibility and permeability from fluid diffusivity during the pre-closure phase. The concept is to analyze fluid leak-off mechanisms during pumping and fracture closure to provide an estimate of reservoir transmissibility. The results from this approach are validated with well performance, wherein transmissibility data generated from the correlation are correlated to the productivity index. The validation tool showed a strong correlation between the reservoir transmissibility and the productivity index, implying the influence of reservoir transmissibility on fluid deliverability.

Methodology

The main purpose of this paper is to introduce a simplified approach to predict reservoir transmissibility and permeability from DFIT closure data in the absence of radial flow in the formation. The data obtained by performing CA and ACA are related through the fluid leak-off mechanism. This idea is built on the observations revealing different fluid leak-off efficiencies for different reservoir transmissibility values. Thus, the proposed methodology provides a correlation derived based on fluid leak-off to estimate reservoir transmissibility and permeability. The methodology begins with retrieving CA and ACA data from offset wells targeting the same formation. The correlation is then derived after examining the relationship between fluid leak-off efficiency from CA and reservoir transmissibility from ACA.

The results from the proposed methodology will assist in estimating reservoir permeability in the absence of ACA data, mainly due to the excessive DFIT monitoring time required to reach radial flow in the formation.

Closure Analysis: Fluid Leak-off

Fluid leak-off upon the generation of a hydraulic fracture is governed by the leak-off coefficient equations that Settari, A. (1985) classified based on three fluid loss mechanisms: fluid loss due to filter cake, filtrate viscosity, and reservoir fluids, each presented in the following **Equations (1), (2), and (3)** respectively:

$$C_w = \sqrt{\frac{k_{cake} \alpha \Delta P_{cake}}{2\mu_{fil}}} \quad (1)$$

$$C_v = \sqrt{\frac{k_{fil} \phi \Delta P_v}{2\mu_{fil}}} \quad (2)$$

$$C_c = \sqrt{\frac{k_r \phi c_t}{\pi \mu_r}} \Delta P_c \quad (3)$$

where C_w , C_v , and C_c are the loss coefficients due to filter cake, filtrate viscosity, and reservoir fluids, respectively. Each fluid loss coefficient contains parameters influencing the leak-off through the subject interval and describing the leak-off mechanism.

Since clean fluid is injected during DFIT, there is no polymer to form a filter cake layer on the fracture wall. Hence, fluid leak-off to the formation is controlled by filtrate viscosity and reservoir fluids losses for a DFIT case. The pressure difference between Instantaneous Shut-in Pressure (ISIP) and fracture closure is the mechanism that governs fluid loss due to resistance of filtrate viscosity in the invaded zone. In the post-closure phase, fluid continues to leak-off through the uninvaded zone due to the resistance of reservoir fluids as the pressure transient approaches the reservoir pressure point. The leak-off mechanism through the reservoir is controlled by the differential pressure between the closure pressure and reservoir pressure.

Fluid losses through the filtrate viscosity and reservoir fluids are linked to the fluid leak-off efficiency in the DFIT pumping interval, indicated in **Figure 1**. Fluid leak-off efficiency is defined as the ratio of how much fluid leaked into the formation from the total volume injected while pumping and initiating the small fracture. The fluid efficiency reflects the fluid losses during DFIT pumping, which occurs for a short period when there is an immediate contact of the generated fracture face with the injected fluid. The DFIT fluid leaks through the formation, displacing the formation fluids at the fracture face before leaking off through the invaded zone, generating a water bank around the fracture face. At that moment, the fluid loss coefficient through the reservoir is valid and governed by the following **Equation (4)**:

$$C = \sqrt{\frac{k_r \phi c_t}{\pi \mu_r}} \Delta P_{\text{net}} \quad (4)$$

The fluid leak-off process will continue as the fracture propagates and the new fracture face interacts with injected fluids. During that process, fluid loss into the formation is derived by the net pressure defined as the differential pressure in the fracture (treating pressure) and ISIP.

Closure Analysis: Fluid Leak-off Efficiency

The proposed approach focuses on analyzing the leak-off behavior of pressure transient in the pre-closure phase through the leak-off efficiency to connect it to reservoir permeability. Obtaining the fluid leak-off efficiency from the CA is the first piece solving for the target leak-off efficiency and transmissibility correlation. The approach starts by conducting CA, which requires multiple tools to evaluate the pressure transient data. The first tool is the log-log diagnostic plot, which translates the behavior of pressure change to easily identify the type of flow regimes the well is experiencing. The diagnostic plot has a log-log scale of pressure transient and derivative of pressure transient versus time elapsed after shut-in. **Figure 2** demonstrates a typical log-log diagnostic plot for a DFIT case. The slope of pressure change derivative is used to identify the flow regime to then evaluate the instance of fracture closure.

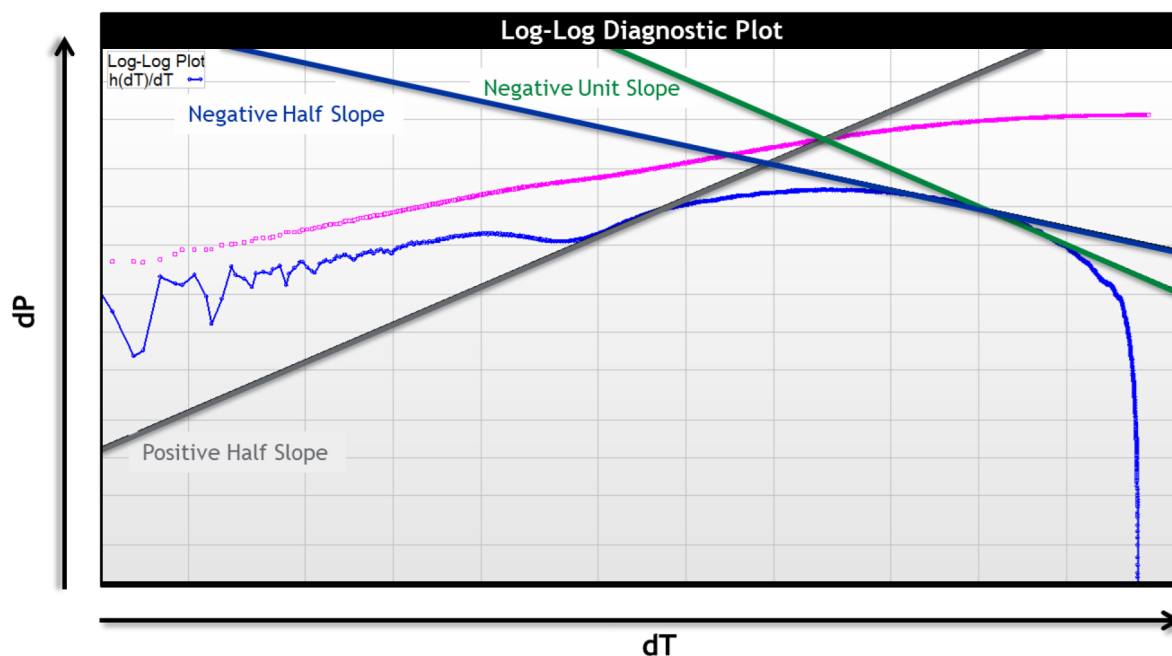


Figure 2—Log-Log Plot with Flow Regimes Identification. Negative Half Slope $h(dT)/dT$ dP Log-Log Diagnostic Plot

For every DFIT case, linear flow to the formation must be detected to have confidence that there is sufficient fracture created to have a representative case. The linear flow is developing at the early time of the pre-closure phase, and it is recognized by a positive half slope.

While fluid is leaking-off into formation and the fracture completely closes, the slope sign changes approaching to negative half slope, representing formation linear flow. Establishing linear flow through the formation confirms fracture closure, and that regime is identified with the negative half slope picked on the log-log plot as captured in [Figure 2](#).

Once closure is determined through the diagnostic plot, a second tool is used to interpret pressure transient data to obtain fracture properties. The most common tools used are the G-function and its derivative plots. [Nolte \(1979\)](#) introduced the G-function which is a dimensionless time function that defines the closure parameters and fluid leak-off behavior. In this procedure, the G-function derivative plot was applied to every DFIT case to avoid any discrepancy when identifying fracture closure using other plots. Since the derivative of the G-function ($G-dP/dG$) detects any deviation of pressure behavior during DFIT monitoring, the fracture closure time can be easily determined. CA process starts by identifying the closure of the fracture generated with DFIT. The signature of fracture closure is recognized as the peak in the $G-dP/dG$ curve, followed by continuous decline approaching stabilization, as illustrated in [Figure 3](#). Closure pressure picking is selective to a high degree if the interpretation of the G-function plot is not covering the entire behavior after the first peak. The pressure transient behavior can exhibit multiple closure signatures, indicating multi-fractures closing at once, or the closure of natural fractures the during DFIT post-injection monitoring. Another indication of false closure is when there is a single closure signature, but the derivative curve goes to negative slope with a magnitude of less than a half unit. In this case, there is a fracture generated in another direction that is not in the dominant direction. For both cases, monitoring time should be extended until a clear signature of fracture closure is detected. With the fracture closure determined, the closure parameters are then obtained, including closure pressure, closure time, leak-off coefficient and fluid efficiency. The fluid leak-off efficiency estimate is computed through the G-function closure time (G_c) using the following [Equation \(5\)](#):

$$Eff(\%) = \frac{G_c}{2 + G_c} \quad (5)$$

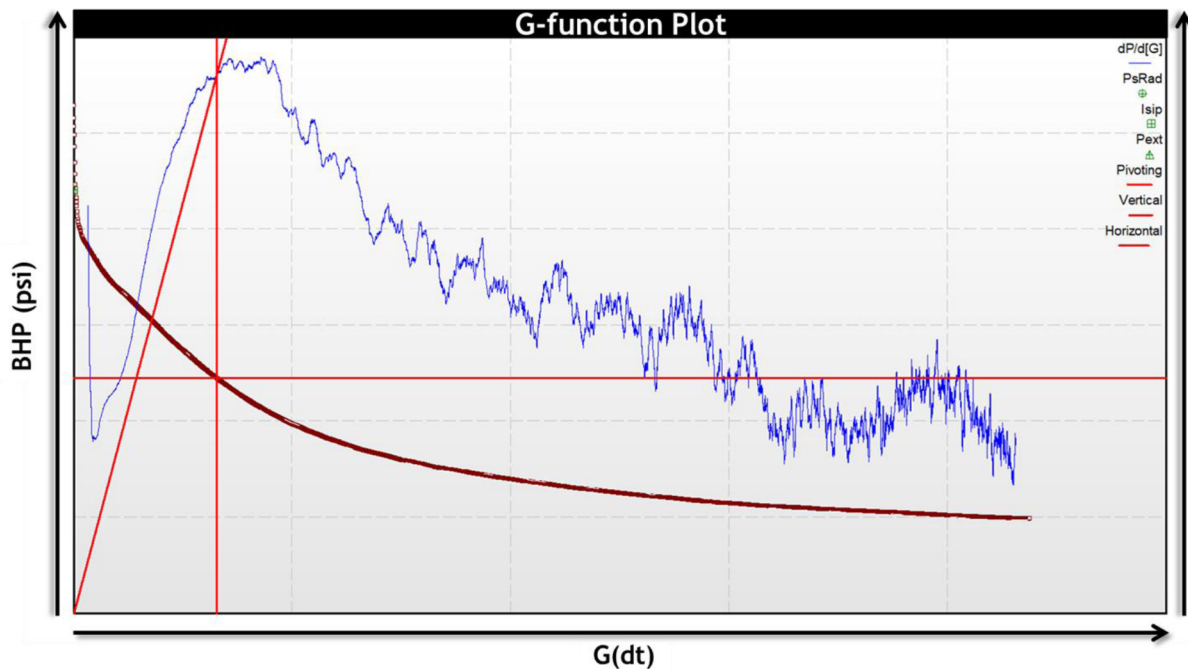


Figure 3—Example of Closure Picking for Closure Analysis.

ACA: Reservoir Transmissibility

Obtaining ACA data is the second piece to solve for leak-off efficiency and transmissibility correlation. Post-closure decline monitoring is resumed until radial flow in the formation is developed. During the post-closure phase, the fluid diffusivity mechanism through the formation is governed by the pressure charge into the uninvaded zone, which is defined as the magnitude of the closure pressure as it approaches reservoir pore pressure. Due to pressure charge during injection, radial flow into the formation starts to appear. Formation radial flow development is dependent on the fluid diffusivity through that formation by displacing existing formation fluids. For extremely tight reservoirs, diffusivity of fluids into formation is low, hence, it requires longer monitoring time to clearly identify radial flow. The radial flow regime is identified using the log-log diagnostic plot by a negative unit slope. For wells that developed radial flow, the ACA is conducted to estimate reservoir parameters from different pressure decline analysis tools such as Horner's plot or the equivalent method of pseudo-radial flow plot (Nolte, 1997).

The reservoir parameters are determined from the slope of the straight line obtained from Horner's plot in the post-closure phase, or the negative unit slope using pseudo-radial flow plot, **Equation (6)**. The pressure differential (ΔP_c) is the difference between closure pressure and the estimated reservoir pressure. The impulse time function $F_R(\Delta t)$ is a dimensionless time function for the pseudo-radial flow for low permeability formations, proposed by Nolte (1997). The plot of pressure differential and impulse time function results in an equation of a straight-line. The slope of the straight-line (m_R) holds key reservoir parameters, including reservoir transmissibility term, **Equation (7)**.

$$\Delta P_c = m_R \times F_R(\Delta t) \quad (6)$$

$$m_R = \frac{\pi \mu V_i}{16 k h t_{CL}} \quad (7)$$

Solving for reservoir transmissibility from the slope equation provides the reservoir term controlling the ACA. To prevent any uncertainties when determining the exact hydraulic fracture height (h), or viscosity of formation fluids (μ), the transmissibility term $\left(\frac{kh}{\mu}\right)$ is used to construct the correlation.

Fluid Leak-off Efficiency and Transmissibility Correlation

The correlation between CA and ACA parameters is established based on the fluid leak-off behavior in the first contact of the injection fluid with the fracture face during DFIT pumping. Leak-off relates to the displacement of formation fluids, and it links the volume of injected fluid leaked off through the formation to the fluid loss coefficient during this phase. As a result, the leak-off efficiency is correlated to the reservoir permeability through the leak-off mechanism during the time of DFIT injection. This approach is developed based on retrieving CA and ACA results from previous DFITs conducted for the same target formation. The fluid leak-off efficiency from CA, and reservoir transmissibility from ACA were collected for multiple wells. Observations revealed that for different fluid leak-off efficiencies, there were different transmissibility value, which strongly correlated with an exponential curve as shown in **Figure 4**.

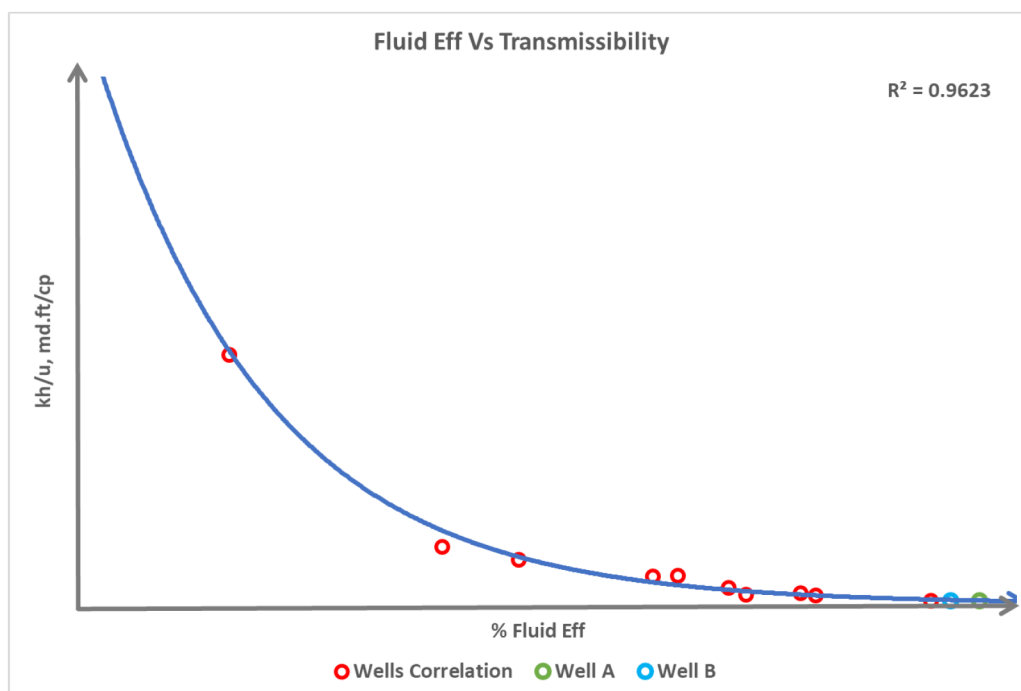


Figure 4—Reservoir Transmissibility versus Fluid Leak-off Efficiency Obtained by the Proposed Methodology.

Consequently, the developed correlation through the field DFIT data proved that fluid leak-off efficiency is an index for formation transmissibility. The fluid leak-off efficiency and the reservoir transmissibility are related through the loss coefficient while pumping. The produced correlation is an imperial equation that is used as a tool to estimate the permeability value when radial flow through the formation is not yet present. Substituting fluid leak-off efficiency value for any subject well into this correlation yields the estimation of reservoir transmissibility with high accuracy. From the estimated transmissibility value, the reservoir permeability can be obtained if reservoir fluid viscosity and height were known. The reservoir fluid viscosity can be estimated from PVT reports of nearby producing wells or from the specific well if samples were available, while height is estimated from fracture modeling simulators. The reservoir permeability is analogous to the transmissibility when using this approach. However, the uncertainty associated with using the permeability correlation will slightly be higher since now the correlation is influenced by the estimates of the fluid viscosity and the fracture height. With the available data to construct the proposed correlation, the data of CA, fluid properties, and rock properties, the reservoir permeability can be directly estimated using the proposed approach.

Results and Discussion

The scope of this paper is to evaluate the reservoir transmissibility/permeability from the CA utilizing previous transmissibility data points from offset wells. The applications of the proposed workflow are for wells targeting extremely tight reservoir where excessive DFIT operational time is required to reach formation radial flow. The basis of this approach was built on the fluid leak-off behavior during DFIT injection, particularly the fluid loss of the immediate contact between fluid injected and the fracture face. The fluid leak-off during pumping is reflected on the fluid leak-off efficiency obtained by CA expressed as the ratio of how much fluid volume leaked-off to formation. The scope of this study focused on DFIT from vertical wells to minimize the uncertainty associated with horizontal lateral sections or potential formation heterogeneity.

The methodology was applied on multiple wells, where CA and ACA data are first retrieved from DFITs conducted on wells targeting the same formation. The values of fluid leak-off efficiency from CA data

points were plotted against the reservoir transmissibility obtained from ACA. The final product of this approach is revealed in **Figure 4**, presenting a correlation between reservoir transmissibility and leak-off efficiency via an inverse exponential function. The data showed a strong correlation, with a Coefficient of Determination value of more than 96%. The final product proved that the fluid leak-off efficiency is an index of the transmissibility. For a reservoir case with low transmissibility, the leak-off efficiency will be high since small fracture fluid volume leaked-off to the formation, indicative of low transmissibility.

The main purpose of developing this methodology was to gain understanding on formation deliverability by evaluating the reservoir transmissibility as a potential factor related to the fluid leak-off efficiency. Understanding the estimated reservoir transmissibility will assist in evaluating the hydraulic fracture job and formation deliverability. This is owing to the permeability being the main parameter in fracture modeling, with fluid leak-off as a deriving factor to various simulation runs. Furthermore, to validate the proposed approach, the outputs from the correlation were confirmed with well performance via productivity index.

Application: Wells A & B Example

The proposed methodology work was applied to predict the reservoir permeability for wells A and B. The subject wells were drilled vertically and both were targeting the same formation located at different depths. DFIT was conducted to study the reservoir characteristics and evaluate the fracture design for the subject wells at different target depths. DFIT operational steps involved pumping treated water enough to initiate a small fracture, followed by the recording of wellhead pressure. The pressure decline was monitored up to closure time where linear flow is confirmed by negative half-slope through a diagnostic plot, and DFIT was concluded before radial flow was developed.

The methodology was applied to construct leak-off efficiency versus reservoir transmissibility correlation. The final product was used to evaluate the reservoir permeability for the subject wells based on the CA executed to get fluid leak-off efficiencies. CA results revealed there are two different fluid leak-off values, which yielded two different reservoir transmissibility values using the empirical correlation, as in **Figure (4)**. The next operation after DFIT was stimulation then flowback both wells to obtain production parameters used to validate the empirical correlation. Based on post-stimulation reports, fluid viscosity and fracture height were used to find the permeability of the two regions. The estimated reservoir transmissibility value was the main key to study the feasibility of sidetracking a horizontal well at one of the tested zones.

Well Performance

Well performance through productivity index is the tool used to validate the proposed methodology. Since transmissibility is the controlling factor on fluid deliverability, the predicted values from the empirical correlation were validated through well performance. The proposed workflow was applied on a list of wells that had DFIT concluded before reaching radial flow. These wells were stimulated with hydraulic fracturing then attended for flowback testing for clean-up, which yielded deliverability data including gas rate and flowing pressure. The flowback parameters were then used to compute the productivity index using the pressure square method. The transmissibility data was plotted against the well productivity index, as shown in **Figure 5**. The plot demonstrated a strong correlation between the two parameters, with extremely high Coefficient of Determination value. This correlation was justified as the productivity index, representing the general diffusivity equation and transmissibility as main terms in the equation. Hence, well performance validated the proposed methodology, where the predicted transmissibility from the correlation is confirmed through the productivity index from flowback testing.

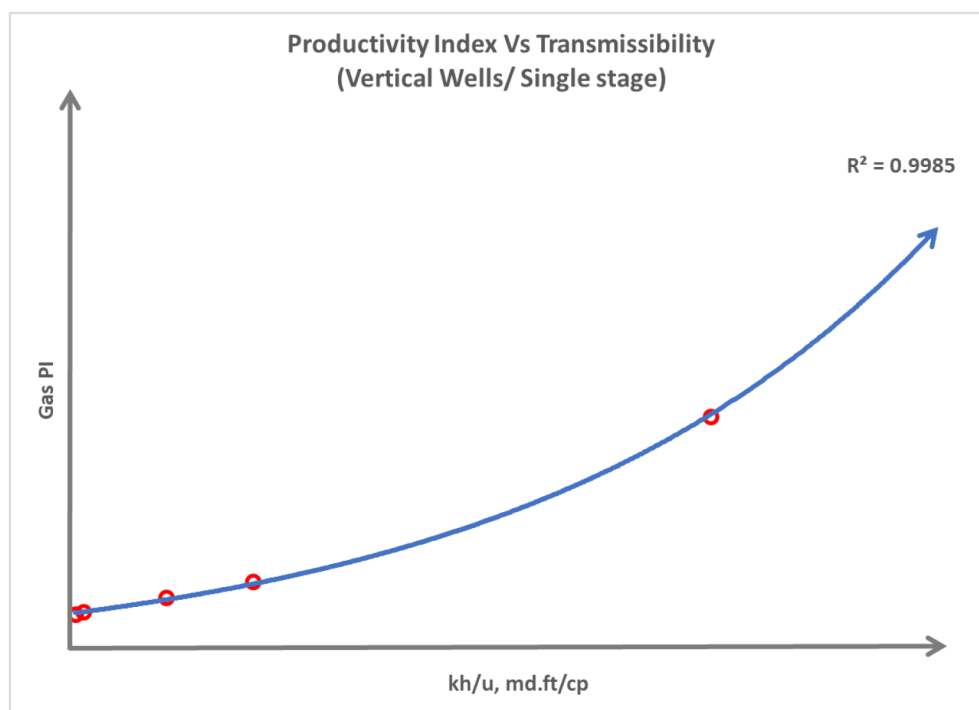


Figure 5—Productivity Index and Reservoir Transmissibility Plot

Conclusion

DFIT is a well testing tool that is widely used to evaluate fracture and reservoir parameters in Unconventional Reservoirs. The reservoir parameters cannot be predicted unless radial flow through the formation is observed. In extremely tight reservoirs, radial flow requires a long time to be developed. This paper proposed a novel approach with simplified steps to estimate the reservoir transmissibility/permeability in the absence of radial flow.

The basis of the proposed methodology is derived on the concept of leak-off mechanisms. Fracture parameters are mostly obtained by G-function, with the main leak-off mechanism controlled by the magnitude of ISIP and closure pressure. The reservoir parameters are acquired from ACA, which require the formation radial flow to be developed. However, transmissibility is also impeded in the leak-off coefficient while pumping DFIT fluid, which is reflected by the leak-off during the immediate contact between the injected fluid and the fracture face. The fluid leak-off efficiency from CA is linked to the reservoir transmissibility through the fluid leak-off coefficient during the DFIT pumping.

The methodology was applied for multiple wells targeting the same formation, where the DFIT results of CA and ACA parameters were retrieved and compiled to derive the correlation. Fluid leak-off efficiency is a part of CA, while the reservoir transmissibility is of ACA. The final product of correlating the leak-off efficiency and reservoir transmissibility is used to estimate the reservoir transmissibility when only fracture closure monitoring is captured. Results revealed that fluid leak-off efficiency is a proxy of formation quality, validated by well performance where productivity index and estimated transmissibility agreed with diffusivity correlation.

Estimating transmissibility/permeability is an essential factor to build and calibrate hydraulic fracture modeling, reservoir modeling, and production modeling. The derived correlation optimizes the DFIT decline monitoring time, which estimates the formation quality when DFIT analysis concluded after reaching closure time. In addition, since the closure estimation is selective, the correlation validates the closure pickings.

Nomenclature

C_w	= Leak-off coefficient through filtercake
C_v	= Leak-off coefficient through invaded zone due to filtrate viscosity
C_c	= Leak-off coefficient through reservoir due to reservoir fluid
C	= Leak-off coefficient during DFIT pumping
k_{cake}	= Permeability of filtercake, mD
k_{fu}	= Permeability of invaded zone, mD
k_r	= Reservoir permeability, mD
P_{BD}	= Breakdown pressure, psi
P_{CL}	= Fracture closure pressure, psi
P_R	= Reservoir pressure, psi
AP_{cake}	= Pressure drop behind filtercake, psi
AP_V	= Pressure drop in the invaded zone, psi
AP_C	= Charge pressure of the reservoir, psi
AP_{net}	= net pressure of the difference between pressure drop at any point and ISIP, psi
a	= cake deposition constant, fraction
ε	= Rock porosity, fraction
η_a	= Fracturing fluid apparent viscosity, cp
η_f	= Filtrate viscosity, cp
η_m	= Formation fluid viscosity, cp
c_t	= Total compressibility, psi ⁻¹
Eff	= Injected fluid leak-off Efficiency, percentage
G_c	= G-function closure time, dimensionless
V_I	= Volume of injected fluid, ft ³
h	= Fracture height, ft
t_{CL}	= Closure time, minute
$F_r(At)$	= Dimensionless time at any point of pressure decline monitoring,

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